UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*2965448557*

**MATHEMATICS (SYLLABUS D)** **4024/01**

Paper 1 **May/June 2007**

**2 hours**

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

**NEITHER ELECTRONIC CALCULATORS NOR MATHEMATICAL TABLES MAY BE USED IN THIS PAPER.**

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

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**NEITHER ELECTRONIC CALCULATORS NOR MATHEMATICAL TABLES MAY BE USED IN THIS PAPER.**

**1** **(a)** Evaluate $3 + 25 \div 2$.

**(b)** Express $17\frac{1}{2}\%$ as a decimal.

*Answer* (*a*) ..... [1]

(*b*) .....[1]

---

**2** Evaluate

**(a)** $\frac{1}{4} + \frac{1}{7}$,

**(b)** $1\frac{7}{8} \div \frac{3}{16}$.

*Answer* (*a*) .....[1]

(*b*) .....[1]

---

**3** It is given that $\frac{2}{3}$, $\frac{8}{d}$ and $\frac{n}{39}$ are equivalent fractions.

Find the value of *d* and the value of *n*.

*Answer* *d* = .....[1]

*n* = .....[1]

---

**4** **(a)** A car decelerates uniformly from 20 m/s to 5 m/s in 25 seconds.
Calculate the retardation.

**(b)** Express 20 metres per second in kilometres per hour.

*Answer* (*a*) ..... $m/s^2$ [1]

(*b*) ..... km/h [1]

---

**5** **(a)** Write the following in order of size, starting with the smallest.

$$\frac{66}{100} \qquad 0.\dot{6} \qquad 0.67 \qquad \frac{666}{1000}$$

*Answer* (*a*) ..... , ..... , ..... , ..... [1]
*smallest*

**(b)** The distance of Saturn from the Sun is 1507 million kilometres.
Express 1507 million in standard form.

*Answer* (*b*) .....[1]

---

**6** **(a)** Express 154 as the product of its prime factors.

**(b)** Find the lowest common multiple of 154 and 49.

*Answer* (*a*) .....[1]

(*b*) .....[1]

---

**7** In the quadrilateral $ABCD$, $\hat{A} = x°$, $\hat{B} = 2x°$, $\hat{C} = 3x°$ and $\hat{D} = 4x°$.

**(a)** Find $x$.

**(b)** Explain why $AB$ is parallel to $DC$.

*Answer* (*a*) $x =$.....[1]

*(b)* .....[1]

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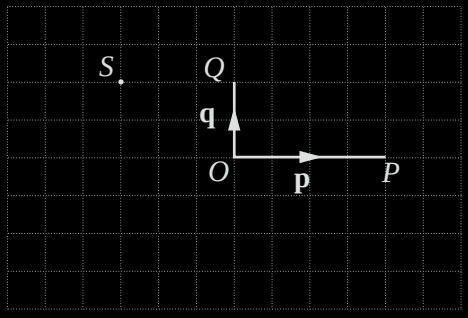
**8** On the grid in the answer space, $\overrightarrow{OP} = \mathbf{p}$ and $\overrightarrow{OQ} = \mathbf{q}$.

**(a)** Given that $\overrightarrow{OR} = \mathbf{p} - \mathbf{q}$, mark the point $R$ clearly on the grid.

**(b)** The point $S$ is shown on the grid.

Given that $\overrightarrow{OS} = \mathbf{q} + h\mathbf{p}$, find $h$.

*Answer* (*a*)



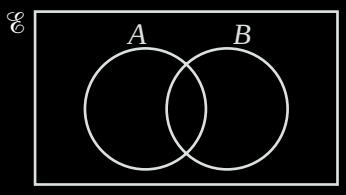

[1]

(*b*) $h =$.....[1]

---

**9** **(a)** The sets $A$ and $B$ are shown on the Venn Diagram in the answer space.
The element $y$ is such that $y \in A$ and $y \notin B$.
On the diagram, write $y$ in the correct region.

*Answer* (*a*)




[1]

**(b)** $\mathscr{E} = \{x : x \text{ is an integer and } 1 \leqslant x \leqslant 8\}$.
$P = \{x : x > 5\}$.
$Q = \{x : x \leqslant 3\}$.

**(i)** Find the value of $n(P \cup Q)$.

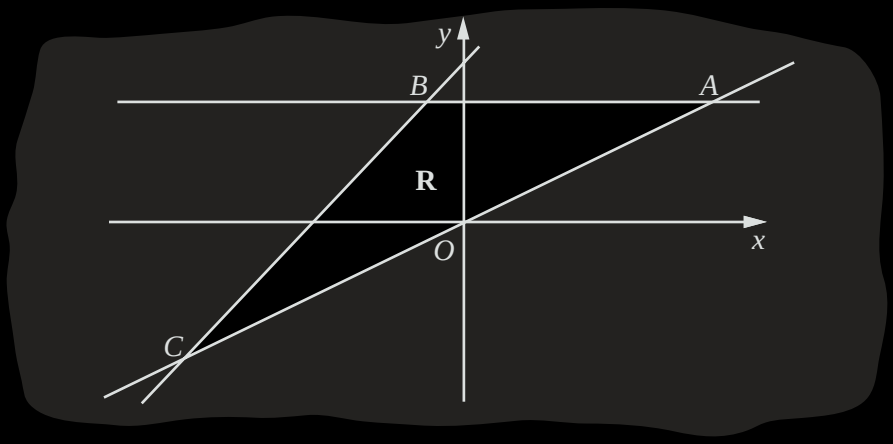
**(ii)** List the elements of $P' \cap Q'$.

*Answer* (*b*)(i) .....[1]

(ii) {.....} [1]

---

**10**




In the diagram, $A$ is the point (6, 3) and $C$ is the point (−8, −4).
The equation of $AB$ is $y = 3$ and the equation of $CB$ is $y = x + 4$.

**(a)** Find the coordinates of $B$.

**(b)** The unshaded region **R** inside triangle $ABC$ is defined by three inequalities.
One of these is $y < x + 4$.

Write down the other two inequalities.

*Answer* (*a*) (..... , .....) [1]

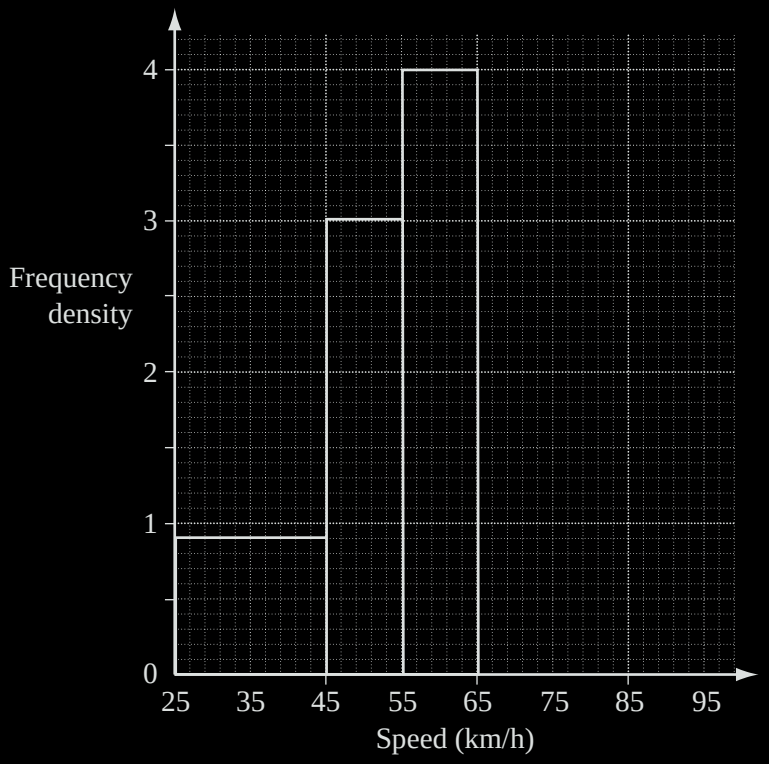
(*b*).....

.....[2]

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**11** On a certain stretch of road, the speeds of some cars were recorded.
The results are summarised in the table.
Part of the corresponding histogram is shown alongside.

| Speed ($x$ km/h) | Frequency |
|---|---|
| $25 < x \leqslant 45$ | $q$ |
| $45 < x \leqslant 55$ | 30 |
| $55 < x \leqslant 65$ | $p$ |
| $65 < x \leqslant 95$ | 12 |




**(a)** Find the value of

**(i)** $p$,

**(ii)** $q$.

*Answer* (*a*)(i) $p =$ .....[1]

(ii) $q =$ .....[1]

**(b)** Complete the histogram. [1]

---

**12** Evaluate

**(a)** $17^0$,

**(b)** $4^{\frac{5}{2}}$,

**(c)** $(0.2)^{-2}$.

*Answer* (*a*) .....[1]

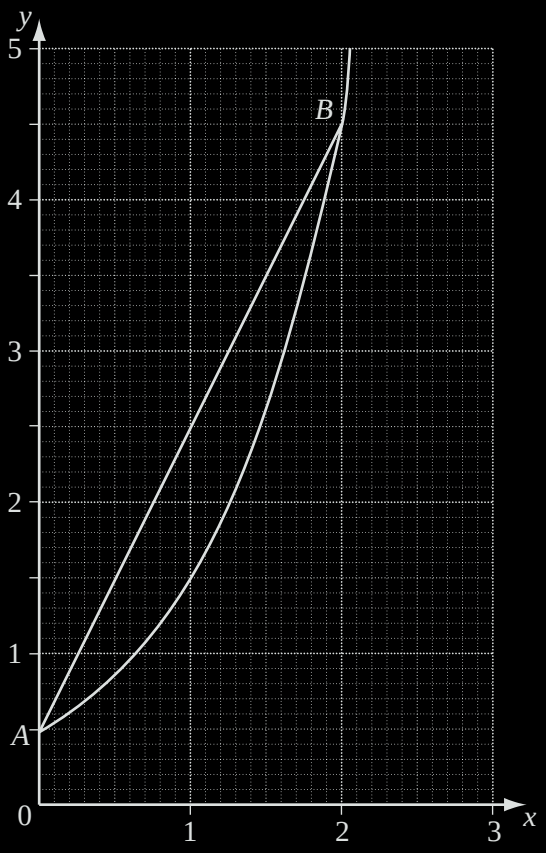
(*b*) .....[1]

(*c*) .....[1]

**13** **(a)** Given that $2y = 3^x$, find $x$ when $y = 40\frac{1}{2}$.

*Answer* (*a*) $x =$ .....[1]

**(b)** The points, $A(0, \frac{1}{2})$ and $B(2, 4\frac{1}{2})$, lie on the curve as shown in the diagram.



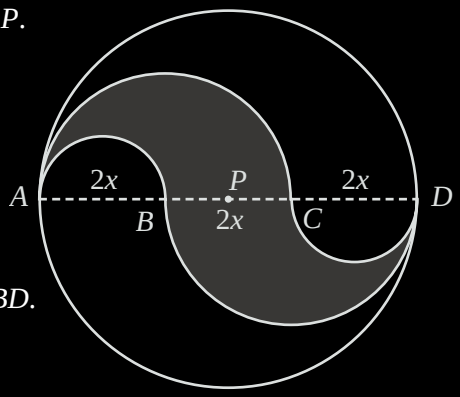

**(i)** Calculate the gradient of the straight line *AB*.

**(ii)** Using the diagram, estimate the value of $x$ at which the gradient of the curve is equal to the gradient of the straight line *AB*.

*Answer* (*b*)(i) ..... [1]

(ii) $x =$ ..... [1]

**14** In the diagram, *ABCD* is a diameter of the circle centre *P*. *AB* = *BC* = *CD* = 2*x* centimetres.




**(a)** Find an expression, in terms of *x* and π, for the circumference of this circle.

**(b)** The perimeter of the shaded region consists of two semicircles whose diameters are *AB* and *CD*, and two semicircles whose diameters are *AC* and *BD*.

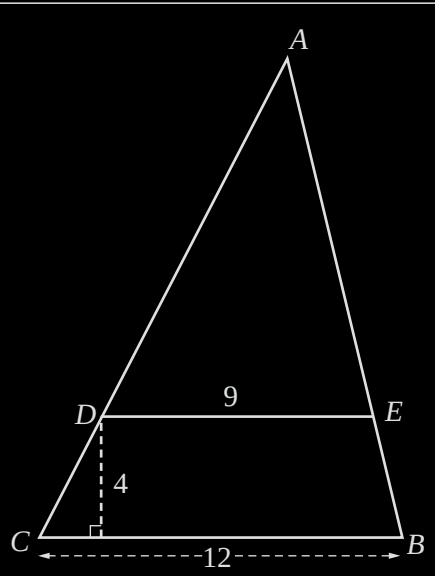
Find an expression, in terms of *x* and π, for the area of the shaded region.

*Answer* (*a*) .....cm [1]

(*b*) ..... $cm^2$ [2]

---

**15** In the diagram, *BCDE* is a trapezium, and the sides *CD* and *BE* are produced to meet at *A*. *CB* = 12 cm, *DE* = 9 cm and the perpendicular distance from *D* to *CB* is 4 cm.




Calculate

**(a)** the area of *BCDE*,

**(b)** the perpendicular distance from *A* to *CB*.

*Answer* (*a*) ..... $cm^2$ [1]

(*b*) .....cm [2]

---

**16** Given that $f(x) = \frac{5x - 4}{3}$, find

**(a)** $f(1\frac{1}{5})$,

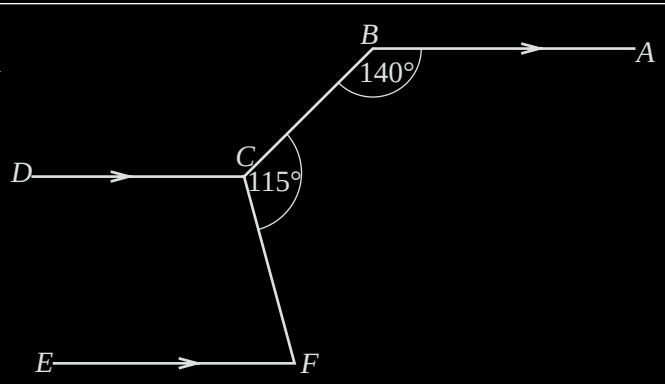
**(b)** $f^{-1}(x)$.

*Answer* (*a*) .....[1]

(*b*) .....[2]

---

**17** In the diagram, the lines *BA*, *DC* and *EF* are parallel.
$A\hat{B}C = 140°$ and $B\hat{C}F = 115°$.




Find

**(a)** $D\hat{C}B$,

**(b)** $D\hat{C}F$,

**(c)** $E\hat{F}C$.

*Answer* (*a*) $D\hat{C}B$ =.....[1]

(*b*) $D\hat{C}F$ = .....[1]

(*c*) $E\hat{F}C$ = .....[1]

---

**18** **(a)** Calculate 5% of \$280 000.

**(b)** A single carton of juice costs \$4.20.
A special offer pack of 3 cartons costs \$9.45.
Ali bought a special offer pack instead of 3 single cartons.

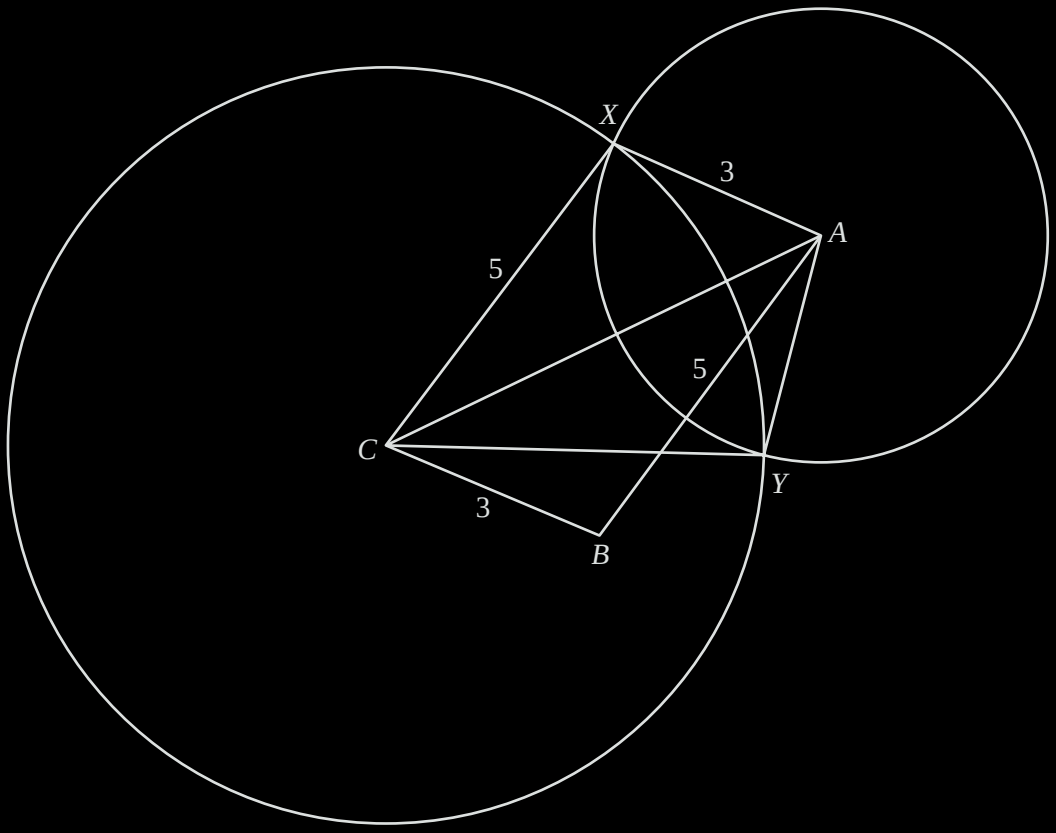
Calculate his percentage saving.

*Answer* (*a*) \$.....[1]

(*b*) ..... % [2]

---

**19**




The diagram shows a circle, centre *C*, of radius 5 cm, and a circle, centre *A*, of radius 3 cm. The circles intersect at *X* and *Y*.
*B* is a point such that *AB* = 5 cm and *BC* = 3 cm.

**(a)** Show that triangles *ABC* and *CYA* are congruent.

*Answer* (*a*) .....

.....

.....

.....[2]

**(b)** Show that the areas of the quadrilaterals *ABCX* and *AYCX* are equal.

*Answer* (*b*) .....

.....

.....

.....[1]

**(c)** State the name of the special quadrilateral *AYCX*.

*Answer* (*c*) ..... [1]

**20** The plan of a field has a scale of 1 cm to 5 metres.

**(a)** Express this scale in the form 1 : *n*.

*Answer* (*a*) .....[1]

**(b)** The plan was made by measuring angles from two points, *A* and *B*, 50 m apart.
The line *AB* is drawn to scale in the answer space below.

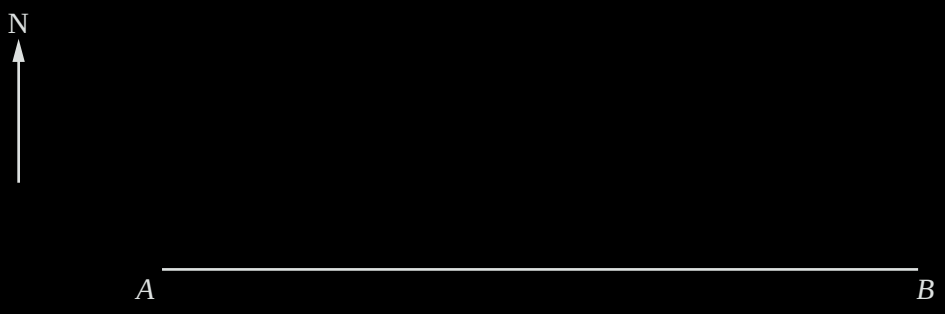
**(i)** A tree is at the point *T* in the field.
$B\hat{A}T = 35°$ and $A\hat{B}T = 70°$.

Locate and label *T* on the plan.

**(ii)** Given that *A* is due west of *B*, state the bearing of *T* from *B*.

**(iii)** By making an appropriate measurement, find the actual distance, in metres, of the tree from *B*.

*Answer* (*b*)(i)




[1]

*Answer* (*b*)(ii) ..... [1]

(iii) ..... m [1]

**21** Factorise

**(a)** $2x^2 - 7x - 15$,

**(b)** $2yt - 8ys - zt + 4zs$.

*Answer* (*a*) .....[2]

(*b*) .....[2]

---

**22** **(a)** Solve

**(i)** $9 - k < 7$,

**(ii)** $\frac{5}{2t} = \frac{1}{12}$.

**(b)** Solve the simultaneous equations $x + y = 29$,
$4x = 95 - 2y$.

*Answer* (*a*) (i) .....[1]

(ii) $t$ = .....[1]

(*b*) $x$ = ..... $y$ = .....[3]

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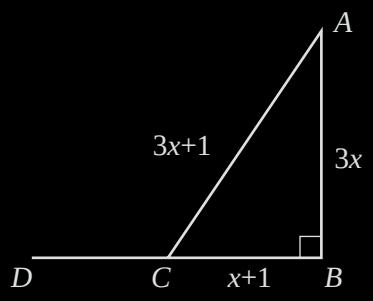
**23** **(a)**

| | sin | cos |
|---|---|---|
| 30° | 0.5 | 0.87 |
| 60° | 0.87 | 0.5 |

Using as much information in the table as necessary, evaluate 2sin150°.

*Answer* (*a*).....[1]

**(b)**




In the triangle $ABC$, $A\hat{B}C = 90°$, $AB = 3x$ cm, $BC = (x + 1)$ cm and $AC = (3x + 1)$ cm.

**(i)** Form an equation in $x$ and show that it reduces to $x^2 - 4x = 0$.

*Answer* (*b*)(i) .....

.....

.....

.....

.....

.....[2]

**(ii)** Find the value of $x$.

**(iii)** Given that $BCD$ is a straight line, state the numerical value of $\cos D\hat{C}A$.

*Answer* (*b*) (ii) $x$ = .....[1]

(iii) .....[1]

**24** **(a)** Evaluate $\begin{pmatrix} 12 \\ 4 \\ 6 \end{pmatrix} - 3\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$.

*Answer* (*a*) $\begin{pmatrix} \quad \\ \quad \\ \quad \end{pmatrix}$ [1]

**(b)** A business makes toy buses and toy lorries.
The following table is used in calculating the cost of making each toy.

| | Labour (hours) | Wood (blocks) | Paint (tins) |
|---|---|---|---|
| Bus | 2 | 3 | 1 |
| Lorry | 1 | *w* | 2 |

Labour costs \$10 per hour, wood costs \$1 per block and paint costs \$*p* per tin.

The information above can be summarised in the matrices **A** and **B**,

where $\mathbf{A} = \begin{pmatrix} 2 & 3 & 1 \\ 1 & w & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 10 \\ 1 \\ p \end{pmatrix}$.

**(i)** Given that $\mathbf{AB} = \begin{pmatrix} 28 \\ 24 \end{pmatrix}$, find

**(a)** *p*,

**(b)** *w*.

**(ii)** Evaluate $(100 \quad 200)\begin{pmatrix} 28 \\ 24 \end{pmatrix}$.

**(iii)** Explain what your answer to **(ii)** represents.

*Answer* (*b*)(i)(*a*) *p* = .....[1]

(*b*) *w* = .....[1]

(ii) .....[1]

(iii).....[1]

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**25** The heights of 40 children were measured.
The results are summarised in the table below.

| Height ($h$ cm) | $105 < h \leqslant 115$ | $115 < h \leqslant 125$ | $125 < h \leqslant 135$ | $135 < h \leqslant 145$ |
|---|---|---|---|---|
| Frequency | 5 | 10 | 20 | 5 |

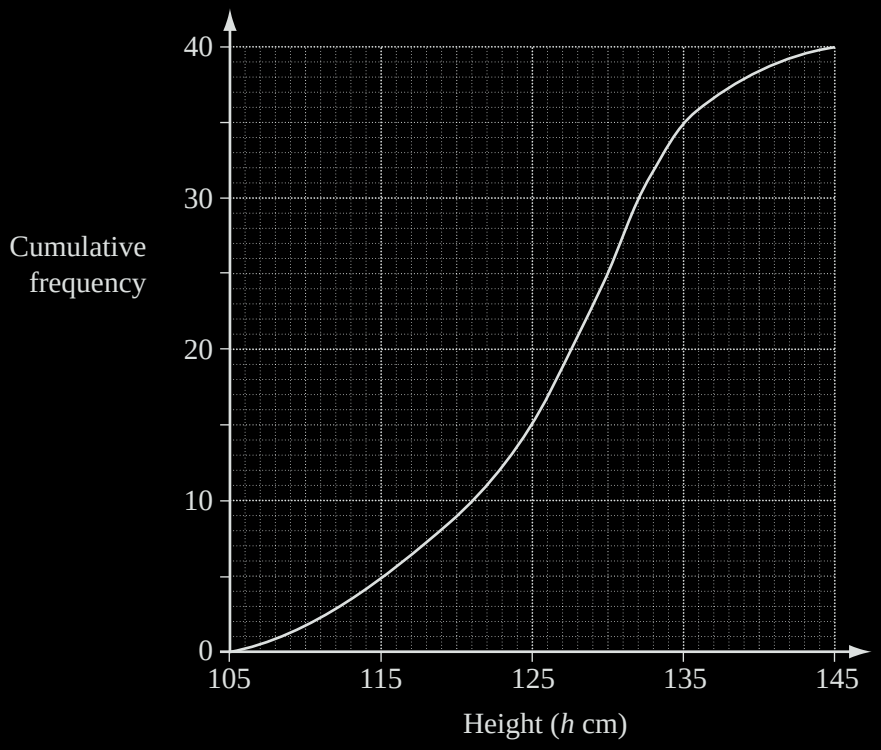
**(a)** **(i)** Identify the modal class.

**(ii)** Calculate an estimate of the mean height.

*Answer* (*a*)(i) .....[1]

(ii) .....cm [3]

**(b)** The cumulative frequency curve representing this information is shown below.




Use the curve to find

**(i)** the interquartile range,

**(ii)** the number of children whose heights are in the range 120 cm to 130 cm.

*Answer* (*b*)(i) .....cm [2]

(ii) .....[1]

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